

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

IBCP308 – Community Project

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PROJECT SYNOPSIS

Predictive Flood Alert System: A Satellite-Based and Machine Learning Approach to Flood Risk Forecasting in Urban Bengaluru

Domain	Space & Earth Observation (secondary relevance: National Importance – Disaster Management)
Team Size	3 students
Course	IBCP308 – Community Project, Dept. of CSE, DSCE
SDG Mapping	SDG 11 (Sustainable Cities and Communities) – Target 11.5; SDG 13 (Climate Action) – Target 13.1

1. Abstract

Bengaluru experiences recurring monsoon flooding and urban waterlogging, driven by unplanned urbanisation, encroachment of natural lakes and storm-water drains, and the loss of natural drainage capacity. Existing civic response largely relies on reactive, post-event damage mapping rather than advance warning. This project proposes a Predictive Flood Alert System that combines satellite-derived hydrological indicators, historical rainfall data, and a machine-learning forecasting model to estimate flood risk for a defined urban drainage basin 24–48 hours in advance, and to communicate that risk to residents and civic stakeholders through an automated alert and a live risk-visualisation map. The system is designed as an end-to-end proof-of-concept pipeline: data ingestion, model-based forecasting, alerting, and visualisation, validated against at least one documented historical flooding event.

2. Introduction

Urban flooding in Bengaluru has intensified over recent decades as a direct consequence of rapid, unplanned urban growth. Studies of the city's land-use change report a concretisation increase of over 1000% between 1973 and 2016–17, accompanied by an 88% decline in vegetation cover and a 79% decline in wetlands [1]. A large share of this loss is concentrated in the city's lake systems: aerial and satellite-based assessments of Bengaluru's lakes have documented encroachment in the majority of studied water bodies, with individual lakes such as Bellandur and Varthur losing hundreds of thousands of square metres of surface area to encroachment and eutrophication [2]. The resulting loss of natural water storage and drainage capacity

directly increases the frequency and severity of waterlogging during the monsoon season, since encroached lake buffers and blocked storm-water drains can no longer accommodate peak rainfall runoff.

Remote sensing has an established track record as a tool for both diagnosing this problem and responding to it. Synthetic Aperture Radar (SAR) imagery from ESA's Sentinel-1 mission has been widely used for flood inundation mapping because, unlike optical satellites, it can acquire usable imagery through the cloud cover that typically accompanies flood events [3][4]. Separately, data-driven forecasting models — particularly Long Short-Term Memory (LSTM) recurrent neural networks — have been shown to model rainfall-runoff relationships with accuracy comparable to, and in some cases exceeding, traditional physically-based hydrological models [5]. This project combines both lines of work: satellite-derived flood-extent data to characterise historical flood behaviour, and an LSTM-based forecasting layer to anticipate risk ahead of a rainfall event, rather than only mapping it afterward.

3. Problem Statement

There is currently no publicly accessible, forward-looking flood risk system for Bengaluru's flood-prone zones that combines satellite observation with predictive modelling. Existing tools are largely retrospective (mapping past flood extent) rather than predictive, leaving residents and local civic bodies with limited lead time to prepare for or respond to an incoming flood event. This project aims to close that gap for a defined pilot area within the city.

4. Objectives

- To collect and process historical rainfall, satellite-derived flood-extent, and terrain elevation data for a selected Bengaluru drainage basin.
- To design and train a machine-learning model (LSTM-based) capable of forecasting flood risk 24–48 hours in advance from rainfall and terrain inputs.
- To build a live risk-visualisation map of the study area, colour-coded by forecast risk level.
- To implement an automated alerting mechanism that notifies stakeholders when forecast risk crosses a defined threshold.
- To validate the forecasting pipeline against at least one documented historical flooding event in the study area.

5. Literature Review

Ramachandra and Aithal (2016) document the scale of unplanned urbanisation in Bengaluru and its consequences for the city's natural drainage and water-storage capacity, establishing the environmental context and urgency for this project [1]. Gopinath et al. (2022) conducted a satellite-imagery-based time-line analysis of 42 prominent Bengaluru lakes and found that 41 had been encroached upon, with a mean physical area reduction of over 10% between 2000 and 2019 [2], reinforcing the choice of a satellite-monitoring approach for the present study.

On the technical side, Islam and Meng (2022) demonstrate an operational approach to rapid urban flood mapping using Sentinel-1 SAR data on the Google Earth Engine platform, evaluating multiple polarisation combinations for accuracy in an urban setting [3]. Moharrami, Javanbakht and Attarchi (2021) similarly present an automated flood-detection workflow using Sentinel-1 imagery on Google Earth Engine, concluding that the approach is well suited to timely flood-inundation mapping to support disaster-time

decision-making [4]. Both studies inform the satellite data-processing component of this project's methodology.

For the forecasting component, Kratzert et al. (2018) introduced the use of LSTM networks for rainfall-runoff modelling, showing that the architecture's ability to learn long-term temporal dependencies makes it well suited to hydrological forecasting tasks, including in settings with limited physical parameterisation data [5]. This forms the methodological basis for the predictive layer proposed in this project, adapted here to an urban storm-water context rather than a river catchment.

6. Methodology

The project methodology is structured into five stages, aligned with the four review milestones of the Community Project framework:

- **Study area & data collection:** Selection of one flood-prone drainage basin in Bengaluru; collection of historical rainfall records (IMD / Karnataka State Natural Disaster Monitoring Centre), Sentinel-1 SAR flood-extent imagery, SRTM digital elevation data, and available storm-water drain layout data.
- **Pre-processing:** Radiometric correction and water-extent classification of SAR imagery; terrain feature extraction from DEM; temporal alignment of rainfall and flood-extent datasets.
- **Model development:** Training an LSTM (or comparable gradient-boosted baseline, e.g. XGBoost) model on historical rainfall-and-terrain features against observed flood extent, to output a per-zone flood-risk score.
- **Live pipeline & alerting:** Integration of a live rainfall-forecast API as model input; a Telegram-bot-based alerting mechanism triggered when forecast risk crosses a defined threshold.
- **Visualisation & validation:** A Leaflet-based interactive map displaying live and historical (replay) risk levels; validation of model output against at least one documented historical flood event in the study area.

7. Sustainable Development Goal (SDG) Mapping

This project is mapped to SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action), consistent with the Space & Earth Observation domain guidance provided in the Community Project framework.

SDG	Target	Relevance to Project
SDG 11 – Sustainable Cities and Communities	Target 11.5: significantly reduce deaths and impacts from disasters, including water-related disasters	The system directly targets early warning for urban flooding, reducing the impact of flood events on residents in the study area.
SDG 13 – Climate Action	Target 13.1: strengthen resilience and adaptive capacity to climate-related hazards	Predictive flood alerting builds adaptive capacity against increasingly intense monsoon rainfall linked to climate variability.

8. Institutional Relevance and Data Sources

The project draws on freely available data and infrastructure from established institutional sources, consistent with the Community Project's emphasis on genuine, verifiable community problems:

- Satellite data: ESA Copernicus Sentinel-1 (SAR) and Sentinel-2, accessed via Google Earth Engine and the Copernicus Open Access Hub.
- Elevation data: NASA/USGS Shuttle Radar Topography Mission (SRTM) digital elevation model.
- Rainfall data: India Meteorological Department (IMD) and the Karnataka State Natural Disaster Monitoring Centre (KSNDMC).
- Prior academic grounding: land-use and lake-encroachment studies from the Centre for Ecological Sciences, Indian Institute of Science (IISc), Bengaluru [1][2].
- Civic context: BBMP/BWSSB storm-water drain and lake records, where publicly available, for cross-referencing model output against documented infrastructure.

9. Expected Outcomes

The completed project will deliver a working proof-of-concept pipeline covering data ingestion, machine-learning-based flood risk forecasting, automated alerting, and map-based visualisation for the selected study area. Model output will be validated against at least one historical flooding event through a replay demonstration. The project explicitly does not claim operational-grade forecasting accuracy comparable to government disaster-management systems; its outcome is a demonstrated, extensible methodology for satellite-and-ML-based urban flood early warning.

10. Team Contribution Plan

Member 1	Data collection & pre-processing: rainfall records, SAR flood-extent classification, DEM feature extraction.
Member 2	Model development: LSTM/XGBoost training, evaluation, and validation against historical events.
Member 3	Systems integration: live rainfall API pipeline, Telegram alerting bot, Leaflet map frontend.

References

- [1] Ramachandra, T.V. and Aithal, B.H., 2016. Bangalore's Reality: towards unlivable status with unplanned urban trajectory. *Current Science (Guest Editorial)*, 110(12), pp.2207–2208.
- [2] Gopinath, R., Akarsh, C.U. and Muralidhar, M.E., 2022. Time-line based aerial analysis for impact of rampant urbanization on lakes of Bengaluru (India). *Civil and Environmental Engineering Reports (CEER)*, 2022 Proceedings.
- [3] Islam, M.T. and Meng, Q., 2022. An exploratory study of Sentinel-1 SAR for rapid urban flood mapping on Google Earth Engine. *International Journal of Applied Earth Observation and Geoinformation*, 113, p.103002.
- [4] Moharrami, M., Javanbakht, M. and Attarchi, S., 2021. Automatic flood detection using Sentinel-1 images on the Google Earth Engine. *Environmental Monitoring and Assessment*, 193(5), pp.1–17.
- [5] Kratzert, F., Klotz, D., Brenner, C., Schulz, K. and Herrnegger, M., 2018. Rainfall–runoff modelling using Long Short-Term Memory (LSTM) networks. *Hydrology and Earth System Sciences*, 22(11), pp.6005–6022.